Tuning the System

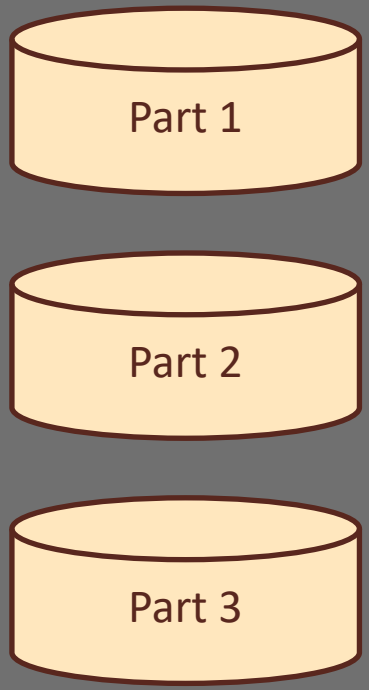
Improve system resources

Not always feasible, but can improve every query.

- Faster CPU
- More physical memory
  - Reduce need for I/O
- Faster I/O
  - SSDs instead of disks
  - Distribute over multiple drives to increase bandwidth

Parallelize data over multiple disks

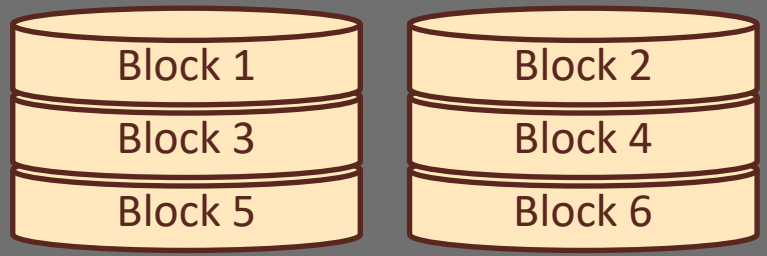
Partitioning



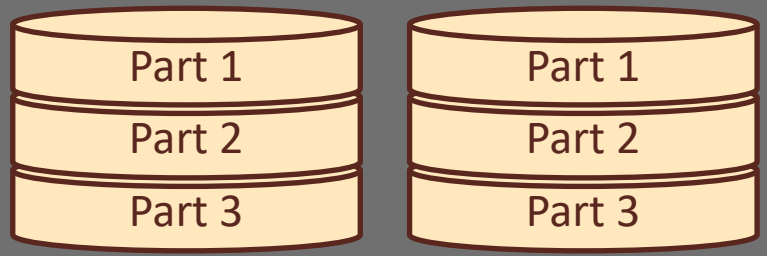
Can partition …

- Different objects
- Different rows of same table
  (Horizontal partitioning = sharding)
- Different columns of same table
  (Vertical partitioning)

Striping = RAID 0



Mirroring = RAID 1



Improve query optimization

- Create indices
- Manage table storage
  - CLUSTER, fillfactor, VACUUM, autovacuum
- Create/manage statistics
  - ANALYZE
- Manage optimizer
  - various flags

Experiment! Optimizers vary, but they're always complex!

Override the query optimizer

Sometimes the query optimizer gets it wrong.

- Force it to use index: SELECT … FROM … WITH INDEX …
- Force join order – join_collapse_limit
- Force subquery order – from_collapse_limit

Not PostgreSQL

However, the query optimizer is usually smarter than you.